BIOT Breath-Sensing Mask - Lung Risk Demo

ESP32 + ENS160 + BME280 breath sensing with an in-browser trained risk model

1. Project Overview

This project is a prototype smart breath-sensing mask. An ESP32 reads two low-cost air-quality sensors, an ENS160 (TVOC, eCO_2) and a BME280 (humidity, temperature), and streams the readings to a web dashboard. The dashboard runs a timed 2-minute breathing test, stops the sensors, averages the readings, and shows **one final risk score** calculated by a logistic-regression model trained on a simulated dataset.

Important: the model is trained on **simulated data** and is a demo of the full pipeline (hardware, data, model, dashboard). It is not a diagnostic tool, is not validated, and must not be used for medical decisions. A consumer ENS160 responds to room air, food, alcohol and hand sanitizer, and cannot detect lung cancer.

2. High-Level Pipeline

```
BME280 + ENS160  ->  ESP32 firmware  ->  USB serial / Bluetooth  ->  Dashboard (browser)
 (I2C sensors)      (L and D lines)        (115200 baud / BLE)        timed 2-min test
                                                                          |
train_model.py -> model_sensors.json -> embedded in HTML -> final risk score
```

Data flows left to right. The model is trained once offline in Python; the dashboard only applies the saved coefficients, so no server, Python or dataset is needed at run time.

3. The Core Components

3.1 ESP32 firmware (sketch)

Reads the BME280 (I2C 0x77) and ENS160 (I2C 0x53) on SDA = GPIO21, SCL = GPIO22, samples at 10 Hz, and prints every line on Serial at 115200 baud and over Bluetooth LE (Nordic UART service, NimBLE). Two line types matter:

- **L line** (10 Hz): `L,ms,HP0,TVOC,HP1,humidity,state`. Raw hot-plate resistances plus TVOC. Used by the original exhale-capture state machine.
- **D line** (1 Hz): `D,ms,TVOC,eCO2,humidity,temp,state`. The only line the dashboard reads. It is sent once a second because the ENS160 updates about once a second.

Set `SIMULATE 0` for real sensors. The ENS160 needs about 3 minutes to warm up after power-on before its readings are meaningful.

3.2 train_model.py - Dataset and Trainer

Generates 6,000 simulated sessions (age, smoking history, TVOC, eCO_2, humidity, temperature, label) shaped to the ranges observed on the real board, saves them to `synthetic_breath_sensors_v3.csv`, fits a logistic regression, and exports the learned numbers to `model_sensors.json`. Only age, smoking history and ln(1+TVOC) are model inputs; eCO_2 is dropped because it is almost perfectly correlated with TVOC.

3.3 model_sensors.json - Trained Model

A small text file holding the model: `coef` (weights), `mean` and `std` (input scaling), `intercept`, `lo` and `hi` (training range used to clamp inputs), and test AUC. Its contents are copied into the dashboard HTML.

3.4 risk_final_v4.html - Dashboard

A single HTML file for Chrome or Edge (Web Serial and Web Bluetooth). It connects over USB or Bluetooth, shows live TVOC, eCO_2, humidity and temperature, and runs the timed test:

- Press **Start 2-min test**; the first 10 s are ignored while sensors settle.
- The remaining readings are averaged; at 120 s the dashboard closes the connection (sensors stop).
- The averages go through the model once and a single final score is shown. A weak-breath warning appears if humidity rose by less than 3 points.
- A Demo data button simulates readings so the whole flow can be tried without hardware.

4. Risk Model

```
x = [age, former_smoker, current_smoker, ln(1 + TVOC)]   (each clamped to the training range)
z = intercept + sum( coef_i * (x_i - mean_i) / std_i )
score = 1 / (1 + e^(-z))
```

Current coefficients on standardized inputs: age 0.49, former smoker 0.19, current smoker 0.54, ln(TVOC) 0.69; intercept -1.04. Held-out AUC is 0.71 for age and smoking alone and 0.76 with TVOC added.

5. Risk Bands

Band	Score	Meaning
Low risk	< 35%	Demo score is low for the entered profile and breath reading
Moderate risk	35% to 65%	Intermediate demo score
High risk	> 65%	High demo score; still not a diagnosis

The cut-offs are arbitrary demo values, and the score is calibrated to the simulated 30% prevalence, not a real-world probability.

6. Serial Line Reference

Field (D line)	Example	Meaning
ms	19562	Milliseconds since boot
TVOC	26	Total volatile organic compounds (ppb)
eCO2	406	Estimated CO2 from the ENS160 (ppm), not a true CO2 measurement
humidity	58.4	BME280 relative humidity (%)
temp	28.5	BME280 temperature (C)
state	1	0 baseline, 1 armed, 2 capture, 3 purge

7. Technology Stack

Layer	Technology
Microcontroller	ESP32 Dev Module, Arduino framework
Sensors	ENS160 (TVOC, eCO2) + BME280 (humidity, temperature) over I2C
Wireless	NimBLE-Arduino, Nordic UART service
Model training	Python, numpy, pandas, scikit-learn (logistic regression)
Dashboard	Single HTML + JavaScript, Web Serial and Web Bluetooth
Data	Simulated CSV, JSON model file

8. Running the System

Flash the firmware

```
1. Arduino IDE: board "ESP32 Dev Module", SIMULATE 0, upload at 115200 if 921600 fails
2. Close the Serial Monitor before using the dashboard (it blocks the port)
```

Retrain (only if the dataset or script changes)

```
pip install numpy pandas scikit-learn
python train_model.py      # writes the CSV and model_sensors.json
# paste model_sensors.json over the  const M={...};  line in the HTML
```

Run a test

```
1. Power the ESP32 and wait at least 3 minutes (ENS160 warm-up)
2. Open risk_final_v4.html in Chrome or Edge -> Connect via USB
3. Enter age and smoking history -> Start 2-min test -> breathe normally into the mask
4. After 120 s the sensors stop and the final score appears
```